



BRINGING RESEARCH TO LIFE

THE NHLBI STRATEGIC VISION

FY 2026–2030



National Heart, Lung,
and Blood Institute

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National Heart, Lung, and Blood Institute

National Institutes of Health

www.nhlbi.nih.gov

ACKNOWLEDGEMENTS

The NHLBI appreciates the extensive input from its staff; the National Heart, Lung, and Blood Advisory Council and Board of External Experts, a working group of the Council; and the heart, lung, blood, and sleep communities in the development of this Strategic Vision.

The NHLBI Strategic Vision was released on October 1, 2025, and aligns with current administration priorities and programmatic goals. It will be implemented through September 31, 2030.



National Heart, Lung,
and Blood Institute

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DIRECTOR'S MESSAGE



Transformative vision propels science and its application to advance health and health care, and for over 75 years, the public has entrusted the National Heart, Lung, and Blood Institute (NHLBI) to support and advance research and its translation to improve lives and save lives for people across our nation and around the world. As a public service science institute, our mission is to simultaneously *see over the horizon* to identify future challenges and opportunities while discovering and applying solutions to our current and newly emergent health and public health challenges.

NHLBI's transformative vision comes from the public we serve — from people and communities from every walk of life who bring unique, diverse, and complementary expertise, insights, and lived experience to inform, inspire, and innovate. In this context, the collaborative Strategic Vision we first developed in 2016 catalyzed and powered advances across heart, lung, blood, and sleep (HLBS) science and medicine, moving the entire biomedical enterprise forward to achieve health and healthcare milestones that appeared as distant goals when we glimpsed them over the horizon 8 years ago.

Your contributions — driven by collaborative community action — energized discovery science and its application to gene therapies so that in 2024, multiple U.S. Food and Drug Administration (FDA)-approved therapeutics now exist to ameliorate sickle cell disease, bringing hope and better health to the individuals and families that have so long born the burden of this transgenerational disorder. Similarly, our collaborative partnerships enabled our swift, agile, multifaceted, and ongoing responses to the COVID-19 pandemic. Discovery science provided us the tools while community-powered implementation efforts translated this science into action to save lives and improve health outcomes in every community, including those bearing the gravest burdens — our most underserved and high-risk populations.

Through forward-thinking collaboration with stakeholders and partners, we revisited our Strategic Vision to update our collective “view” over the horizon and to assess emergent hypersonic technological developments that are exponentially accelerating scientific discoveries and medical advances.

Throughout this update, we have renewed and expanded our commitment to diverse and inclusive partnerships, collaborative community-based research, and the rapid translation of discovery science into public health improvements and frontline, personalized, point-of-care medicine.

The updated NHLBI Strategic Vision better enables us to adapt and leverage emergent science and technology to redress today's most pressing health issues, while helping us anticipate the future with foresight and compass points to guide our actions and facilitate our progress. Although emerging tools like artificial intelligence and machine learning promise to further our understanding of human biology and to accelerate improvements in health and health care, we remain deeply committed to the *ethical application* of all scientific and technological discoveries to ensure health for all, and to ameliorate health disparities, including addressing the social determinants of health (SDOH) that are the very foundations of good health. Well health, health promotion, disease prevention, and treatment are inseparable from SDOH, and some of our greatest challenges now and in the future revolve around whole health and the health of our environment.

As new discoveries are made and our understanding of human biology increases, our Strategic Vision will continue to evolve, serving as evergreen guidance to allow us to adaptively respond to rapidly changing challenges, opportunities, discoveries, and resources. Fundamental biology, implementation science, and community partnerships have been the backbone of NHLBI's research since our founding 75 years ago, and those tenets will continue to power transformative science and its implementation for better health and health care.

A cornerstone of our public service science is the continuous and productive engagement of a broad, diverse, and inclusive circle of partners. Your synergistic and complementary voices provide robust, resilient, and adaptive guidance enabling and empowering swift, efficient, and effective leveraging of unprecedented scientific opportunities to redress HLBS health challenges now and in the generations that follow.

We are deeply grateful for your partnership and for your individual and collective expertise and input as we continue our work together to achieve better health, equitable top-tier health care, and effective and responsive public health that improves lives and saves lives for all people across our country and around our world.

Sincerely,

A handwritten signature in dark ink, appearing to read "Gary H. Gibbons".

Gary H. Gibbons, M.D.
Director, NHLBI

NHLBI STRATEGIC PLAN FRAMEWORK

OVERVIEW OF THE NHLBI

MISSION: The National Heart, Lung, and Blood Institute (NHLBI) provides global leadership for a research, training, and education program to promote the prevention and treatment of heart, lung, and blood diseases and sleep disorders to enhance the health of all individuals so that they can live longer and more fulfilling lives.

NHLBI STRATEGIC GOALS

UNDERSTAND HUMAN BIOLOGY

To expand knowledge of the molecular, cellular, and physiological mechanisms governing the normal function of HLBS systems as essential elements for sustaining human health.

To enable and develop a diverse biomedical workforce equipped with the essential research resources to pursue emerging opportunities in science.

DEVELOP WORKFORCE AND RESOURCES

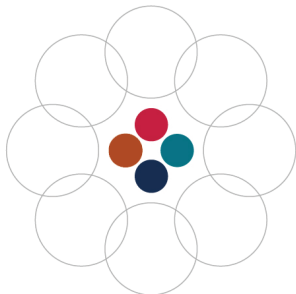
REDUCE HUMAN DISEASE

To extend our knowledge of the pathobiology of HLBS disorders and enable clinical investigations that advance the prediction, prevention, preemption, treatment, and cures of human diseases.

To facilitate innovation and accelerate research translation across the entire research spectrum, bridging basic to clinical, clinical to practice, and population to health impact.

ADVANCE TRANSLATIONAL RESEARCH

STRATEGIC OBJECTIVES



- 1 Understand normal biological function and resilience
- 2 Investigate newly discovered pathobiological mechanisms important to the onset and progression of heart, lung, and blood diseases and sleep disorders
- 3 Investigate factors that account for differences in health status among populations
- 4 Identify factors that account for individual differences in pathobiology and in response to treatments
- 5 Develop and optimize novel diagnostic and therapeutic strategies to prevent, treat, and cure heart, lung, and blood diseases and sleep disorders
- 6 Optimize clinical and implementation research to improve health and to reduce disease
- 7 Leverage emerging opportunities in data science to open new frontiers in research on heart, lung, and blood diseases and sleep disorders
- 8 Further develop and sustain a scientific workforce capable of accomplishing the NHLBI mission

CROSSCUTTING THEMES



Harnessing **data science** and **new technologies** to drive scientific discovery and precision health



Using novel approaches for **addressing health disparities** and tackling their biological underpinnings for heart, lung, and blood diseases and sleep disorders



Furthering the science on the **importance of lifestyle behaviors**



Leveraging the power of **community and patient engagement**



Supporting **women's health** through the lifespan

OVERVIEW

MISSION

The National Heart, Lung, and Blood Institute (NHLBI) provides global leadership for a research, training, and education program to promote the prevention and treatment of heart, lung, and blood diseases and sleep (HLBS) disorders and enhance the health of all individuals so that they can live longer and more fulfilling lives. The NHLBI stimulates basic discoveries about the causes of disease, enables the translation of basic discoveries into clinical practice, fosters training and mentoring of emerging scientists and physicians, and communicates research advances to the public.

Partnerships are imperative to advancing the Institute's mission. The NHLBI creates and supports a robust, collaborative research infrastructure in partnership with private, public, and community organizations, in addition to academic institutions, industry, and other government agencies. Collaborations with patients, families, healthcare professionals, scientists, professional societies, patient advocacy groups, community organizations, and the media are essential to promoting the application of research results and leveraging resources to address public health needs. The NHLBI also fosters ongoing relationships with international organizations to help reduce the burden of HLBS diseases and disorders worldwide.



NHLBI'S ENDURING PRINCIPLES

- Value investigator-initiated fundamental discovery science
- Maintain a balanced cross-disciplinary portfolio (basic, translational, clinical, population science)
- Train and nurture a diverse new generation of leaders for the biomedical workforce
- Support implementation science that empowers patients and enables partners to improve health
- Innovate an evidence-based elimination of health inequities in the United States and around the world



RELEVANT MANDATES

In June 1948, President Harry S. Truman signed the National Heart Act (Public Law 80-655), which created and established the National Heart Institute (NHI). That August, Surgeon General Leonard A. Scheele, by General Circular No. 36, Organization Order No. 14, established the NHI as one of the Institutes at the National Institutes of Health (NIH). On November 10, 1969, the NHI was renamed the National Heart and Lung Institute (NHLI) to reflect the expansion of the NHI's functions. Three years later, the National Heart, Blood Vessel, Lung, and Blood Act of 1972 (P.L. 92-423) expanded the Institute's authority to advance the national response against

heart, blood vessel, lung, and blood diseases. The act provided for intensified and coordinated Institute activities in accordance with a comprehensive, specified National Heart, Blood Vessel, Lung, and Blood Disease Program to be planned by the director and the Advisory Council.

In June 1976, Title I of the Health Research and Health Services Amendments of 1976 (P.L. 94-278) redesignated the NHLI as the NHLBI to even further reinforce the national fight against heart, blood vessel, lung, and blood diseases. It authorized the NHLBI to conduct research in the use of blood and blood products and in the management of blood resources. In June 1993, the NIH Revitalization Act of 1993 (P.L. 103-43) established the National Center on Sleep Disorders Research (NCSDR) within the NHLBI. The NHLBI currently operates under the Research and Investigation Section 301, U.S. Code Citation 428241; National Heart, Lung, and Blood Institute Section 401(a), U.S. Code Citation 42281. The NHLBI director and the National Heart, Lung, and Blood Advisory Council continue to plan the national program under the basic P.L. 92-423 provisions with some refinements.

ORGANIZATION

An organization chart can be found in [appendix I](#).

THE BURDEN OF HEART, LUNG, BLOOD, AND SLEEP DISORDERS

Since its inception, the NHLBI community has fostered a legacy of research excellence through groundbreaking fundamental discovery science, landmark clinical trials and population-based cohort studies, and innovative health education and dissemination efforts. Despite the significant advances in treatment and prevention made during these decades, the chronic conditions resulting from HLBS diseases and disorders still pose a major health and economic burden on the United States. Through its broad research portfolio, the NHLBI is tasked with combating not only the major causes of death and disability, but also rare disorders such as sickle cell disease, thalassemia, and pulmonary fibrosis. Although these disorders are rare in the overall population, the burdens they place on those affected are enormous. Despite research advances, HLBS diseases and disorders continue to be among the leading causes of death and disability in the United States.

 Heart disease and hypertension are among the most common, costly, and preventable health conditions in the United States.¹	 Despite progress in treating and preventing complications of numerous blood diseases, many people still face challenges and a diminished quality of life.²
 Obstructive pulmonary diseases remain a leading cause of death and disability in the United States.²	 Sleep dysfunction and insufficient sleep affect millions of Americans and are associated with cognitive impairments.²

Many of these diseases have a disproportionate impact on people of color, people living in rural communities, and other underserved populations.²

Source: (1) [NIH.gov](https://www.nih.gov), (2) NHLBI ADVANCING Heart, Lung, Blood, and Sleep Research

BUILDING ON PAST SUCCESS TO PREPARE FOR THE FUTURE

The NHLBI must anticipate and capitalize on emerging scientific opportunities as well as foresee and identify approaches to overcome new barriers to progress. Strategic visioning is a process that was initiated in recognition of these needs and in acknowledgment of the transformational scientific advances that are emerging at an accelerating pace.

Investigator-initiated ideas are the principal way we support science, but the complex and nuanced nature of emerging threats to public health require creative solutions that help address problems that are complex, highly evolving, and that have undefined endpoints. This unique approach is evident in NHLBI's responses to the burden of sickle cell disease and to the COVID-19 pandemic.



The **Cure Sickle Cell initiative** was created by the NHLBI in 2018 to use emerging safe and promising genetic therapies to improve the lives of individuals living with sickle cell disease. This initiative actively engages the sickle cell disease community — patients, family members, caregivers, and advocates — to work together on a path to a cure. Cure Sickle Cell encourages collaboration among researchers, industry, nonprofit organizations, and policy-making agencies, all of whom play a role in exploring the scientific questions needed to safely and effectively advance potential therapies.

In response to the COVID-19 pandemic, the NHLBI evolved a strategy not unlike that of the Cure Sickle Cell initiative. In June 2020, the NHLBI developed the Collaborating Network of Networks for Evaluating COVID-19 and Therapeutic Strategies (CONNECTS) initiative. This program facilitated collaboration between experts in pulmonary, critical care, and cardiothoracic surgeries and enabled the development of a master protocol for research on treatments to reduce serious illness and death in patients with COVID-19. In 2020, the NHLBI initiated three clinical trials that later became part of CONNECTS and the NIH Accelerating COVID-19 Therapeutic Interventions and Vaccines (ACTIV) public-private partnership. The goal was to develop a coordinated research strategy for prioritizing and speeding the development of the most promising treatments and vaccines. Known as ACTIV-4, this program investigated the efficacy of anticoagulation and antithrombotic therapy in reducing life-threatening thrombotic cardiovascular and pulmonary complications in newly diagnosed COVID-19 patients not requiring hospital admission, COVID-19 patients hospitalized with moderate and severe disease, and patients recovering from COVID-19.

The NHLBI has either expanded or enhanced innovative and collaborative approaches to other complex challenges to HLBS health. Some notable examples include:

PROJECTS	
The Alliance for Community Engagement – Partnership for Action Toward Health (ACE-PATH)	ACE-PATH seeks to empower communities cumulatively and disproportionately affected by social determinants of health (SDOH) and health disparities across the United States. Its primary goal is to maximize community engagement to achieve optimal health despite climate related risks. Supported community-engaged research projects are charged to (1) Identify, develop, and test effective community-engaged strategies to improve the adoption and use of specific treatments or tools to respond to community health challenges. (2) Support community-engaged research to enhance the adoption, integration, and sustainment of interventions addressing the impacts of local health stressors. (3) Build trust and strong partnerships with all interested parties to exchange knowledge and encourage implementation of promising practices to strengthen and improve community health. (4) Inform strategies to reduce the effects of prioritized local health stressors in U.S. communities.
Accelerating Medicines Partnership® (AMP®)	AMP® is a public–private partnership between the NIH, the U.S. Food and Drug Administration (FDA), multiple biopharmaceutical and life science companies, and nonprofit and other organizations. It aims to transform the current model for developing new diagnostics and treatments. Thus far (2024), AMP has supported nine scientific projects, one of which the NHLBI sponsors, which addresses heart failure.
Community Engagement Alliance (CEAL)	CEAL’s mission is to improve health outcomes, and strengthen community partnerships through community-engaged research that addresses health disparities. The alliance includes various initiatives that promote community-engaged research aimed at improving the health and well-being of communities that most often experience health disparities. These initiatives encourage community-driven, intervention-based research to address disparities in the context of SDOH, as well as the Community Engagement Alliance Consultative Resource (CEACR), a unique program that provides targeted consultations to researchers looking to improve their community engagement and participation efforts.
Precision Medicine and Data Science Programs: BioData Catalyst and Trans-Omics for Precision Medicine (TOPMed)	NHLBI’s TOPMed contributes to NIH’s precision medicine enterprise through the integration of whole-genome sequencing and other omics data (e.g., metabolic profiles, epigenomics, protein and RNA expression patterns) with behavioral, imaging, environmental, and clinical data. NHLBI’s Biodata Catalyst is a cloud-based ecosystem that offers researchers data, analytic tools, applications, and workflows in secure workspaces. It increases access to NHLBI datasets and provides innovative data analysis capabilities to help researchers accelerate discoveries and develop new diagnostic tools, therapeutics, and prevention strategies.
NIH-Global Alliance for Chronic Diseases (GACD) Partnership	This global partnership, with NIH participation, supports implementation and multimorbidity research to reduce the noncommunicable disease burden in low- and middle-income countries, as well as for the first time in American Indian and Alaska Native populations. High-income partner countries also will focus on their Indigenous populations. This program seeks to overcome barriers to implementing evidence-based innovations, tools, policies, strategies, and guidelines.

PROJECTS	
Helping to End Addiction Long-term® (NIH HEAL Initiative®)	HEAL is an aggressive NIH-wide, interagency effort to speed up scientific solutions that stem the national opioid public health crisis. The NHLBI, as part of the HEAL initiative, explores our current understanding of pain and works to establish biomarkers for pain and to develop non-opioid therapeutics for pain management.
NIH Implementing a Maternal health and PRegnancy Outcomes Vision for Everyone (IMPROVE)	An NIH-wide initiative, IMPROVE aims to reduce preventable causes of maternal morbidity and mortality and to improve health for women before, during, and after delivery. IMPROVE supports the IMPROVE Community Implementation Program (IMPROVE-CIP) in partnership with the NHLBI and focuses on decreasing maternal health disparities related to mental health, substance use, psychosocial factors, and social and structural determinants of health. IMPROVE-CIP uses implementation science to bring known effective maternal health interventions into communities severely affected by maternal health disparities.
Maternal Health Community Implementation Program (MH-CIP)	The MH-CIP program aims to reduce maternal deaths and improve maternal health outcomes, especially in populations that are disproportionately affected by high rates of pregnancy-related complications and deaths. MH-CIP develops and tests community-based implementation strategies to increase the adoption, uptake, scale-up, and scale-out of evidence-based interventions that improve health before, during, and after pregnancy. The program emphasizes community-engaged implementation research that is firmly connected to and embedded in affected communities.
Nulliparous Pregnancy Outcomes Study: Monitoring Mothers-to-Be Heart Healthy Study (nuMOM2b HHS)	Funded by the NHLBI, this study examines the effects of pregnancy complications on future cardiovascular health, including social factors that correlate with the future heart health of new parents, especially in communities of color.
Risk Underlying Rural Areas Longitudinal (RURAL) cohort study	Established in 2019, this study is addressing the high burden of chronic heart and lung disease in America's heartland. RURAL is monitoring individuals in 10 rural communities in Alabama, Kentucky, Louisiana, and Mississippi by using community-engaged partners and high-tech mobile research units to collect data including medical histories, familial, lifestyle, and behavioral factors.
Multi-ethnic Observational Study in American Asian and Pacific islander Communities (MOSAIC) initiative	The MOSAIC initiative is a seven-year study, launched in 2023, that focuses on studying cardiovascular health, lung health, mental health, and social determinants of health in individuals that self-identify as having ancestral background from East Asia, South Asia, Southeast Asia; or self-identify as Native Hawaiian and/or Pacific Islander.
Early Intervention to Promote Cardiovascular Health of Mothers and Children (ENRICH) Initiative	The ENRICH initiative is a seven-year community-based, parallel group, two arm, cluster-randomized trial that focuses on enriching evidence-based maternal and child health Home Visiting (HV) with cardiovascular health promotion content (ENRICHed HV) and compares it with usual HV through partnership, collaboration and engagement of HV agencies. About 3,200 parent/child dyads, women less than 34 weeks pregnant, are targeted across 7 academic institutions and about 90 HV agencies.

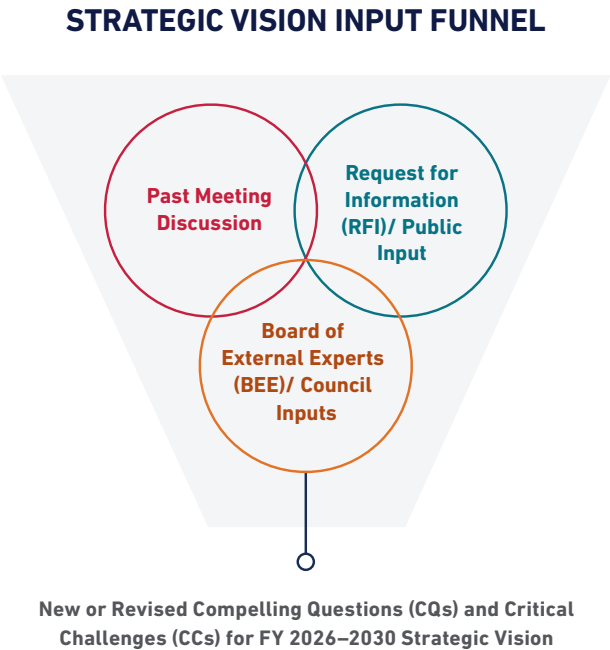
NHLBI has demonstrated a consistent commitment to accomplishing its mission. As we march into the future, we will use innovative problem-solving strategies to address the burden of heart, lung, and blood diseases and sleep disorders.

STRATEGIC VISION
PLANNING PROCESS

Strategic visioning is a process that was initiated in recognition of the transformational scientific advances that are emerging at an accelerating pace. By design, the NHLBI Strategic Vision is a dynamic document, regularly updated with input from people at the leading edge of health and science. Many research goals that once seemed remote are now within reach, and innovative technologies have rendered mundane scientific processes more facile. For example, in the next decade, preventable cardiovascular disease deaths for all age groups could decline by more than 30 percent with greater adoption of proven prevention and treatment strategies. A stroke-free era for children worldwide with sickle cell disease is now imaginable. The NHLBI Strategic Vision is the culmination of an unprecedented effort to engage communities in identifying and prioritizing the most compelling science for which the NHLBI provides leadership.

Over the next 5 years, these collective ideas could catalyze the closing of large gaps in knowledge and could break through significant barriers to research progress in adherence to the Institute’s enduring principles.

This document reflects an update of the 2016 Strategic Vision, which included more than 100 compelling questions and critical challenges that aligned with eight scientific objectives and four goals. For this update, the NHLBI harnessed collective inputs and community-level participation to highlight areas of emerging needs and scientific advancements.





Focus areas were identified following an extensive review of past meetings and discussions by NHLBI leadership, including the National Heart, Lung, and Blood Advisory Council, the Board of External Experts, and a working group of council and division leaders. Input from constituents — scientists, medical professionals, policymakers, patients, patient advocates, and professional groups — was solicited through a series of webinars and a request for information (RFI) that received 141 individual responses and over 700 answers to prompted questions on various topics. These included harnessing data science and new technologies to drive scientific discovery and precision health, using novel approaches for addressing health disparities and tackling their biological underpinnings for HLBS diseases and disorders, leveraging the power of community and patient engagement, furthering the science on the importance of lifestyle behaviors,

supporting women's health through the lifespan, and addressing and reducing the impact of "place" (geography, climate, rural/urban, neighborhood) on HLBS health.

Input sought included perspectives on critical research needs or compelling questions for any of the focus areas, perspectives on challenges or barriers that needed to be addressed to support progress, and the relevance of the eight objectives for the NHLBI Strategic Vision and any critical challenges or questions that were not already incorporated in the NHLBI Strategic Vision. Feedback that resonated as high priorities for the Institute based on timeliness, feasibility, and potential to advance the field of study was incorporated into the updated Strategic Vision.

IMPLEMENTATION OF THE STRATEGIC PLAN

The NHLBI continually assesses and monitors progress toward the previously described four goals and eight objectives while responding to new challenges and building on emerging discoveries. We recognize fully that the Strategic Vision alone is insufficient to affect change without meaningful oversight and implementation.

To confirm that NHLBI-funded activities remain in alignment with the Institute's priorities, periodic analyses are conducted to evaluate research portfolios for potential gaps and opportunities in meeting the goals and objectives of the Strategic Vision. In addition, each division engages publicly through workshops and other efforts to garner inputs and insights supporting the establishment priorities for science. The divisions are assessed on an ongoing basis regarding their oversight of scientific programs and portfolios to achieve these objectives and to ensure that ideas and operations align with the Strategic Vision.

NHLBI leadership engages frequently and meaningfully in cross-disciplinary collaboration to ensure that NHLBI priorities remain dynamic and responsive and that decisions about scientific programs and policies are informed by scientific evidence and input from the scientific community. Every year the Advisory Council convenes with the Board of External Experts to discuss progress made and to keep a pulse on priorities that the NHLBI can support and barriers it can address. Past meetings have included discussions on emerging topics of interest, including the impact of climate, data science, and health disparities. Throughout the year the director, Dr. Gary Gibbons, routinely participates in listening sessions to hear from scientists, patient advocates, and community health leaders across the globe and to discuss health concerns and research opportunities. These encounters serve to identify emerging opportunities and to obtain expert advice and guidance on the implementation of the Strategic Vision.



STRATEGIC PRIORITIES

NHLBI CAPACITY AND OPERATIONAL PRIORITIES

The NHLBI is committed to excellence in its stewardship of the trust of the American public and of the taxpayer funds appropriated by Congress. This commitment aims to ensure that all research and administrative resources are used responsibly to advance the mission to prevent and treat heart, lung, and blood diseases and sleep disorders.

Capacity Priorities

WORKFORCE DEVELOPMENT

To support the staff working at NIH, the NHLBI employs internal programs and training opportunities to promote the professional growth and development of its workforce. These programs aim to foster an appreciation for and commitment to workforce excellence and cultural competency, as well as promote the professional growth of individuals within the Institute. The NHLBI also supports NIH-funded programs that attract, recruit, and retain researchers from varying backgrounds throughout all stages of their careers.

Recruit and Retain

A commitment to promoting scientific opportunities to all groups equally begins with investing in the Institute's workforce through planned and thoughtful recruitment and retention efforts.

Strategic Objective 8 is focused on the development of a scientific workforce that is capable of accomplishing the NHLBI mission. However, capacity and operational priorities are integral to the identity of the NHLBI and are represented throughout the Strategic Vision's goals and objectives.

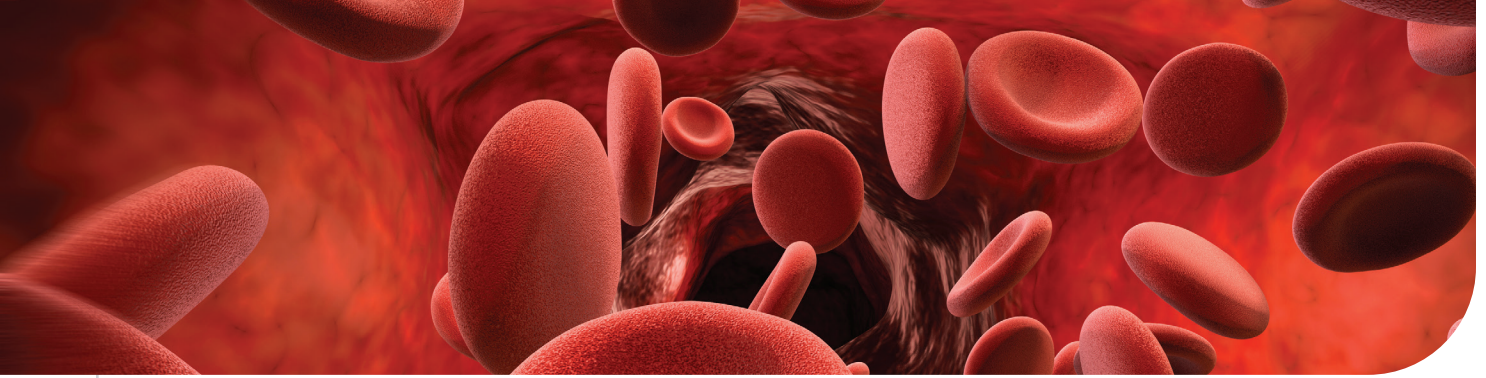


The NHLBI has numerous programs and initiatives intended to encourage individuals to either join the research workforce or further the career they have started. These programs include opportunities for summer research experiences for high school students, which allow a chance for trainees, including those from disadvantaged communities, to hear and get excited about a public service career in the biomedical sciences. For those already focused on the sciences, the Helena Mishoe Fellowship supports postbaccalaureate individuals in biomedical research. Additional NHLBI training programs help junior faculty and transitioning postdoctoral candidates launch their academic research careers and

navigate the complex landscape of academic career progression. The NHLBI has also established re-entry programs for established investigators who have had their research careers interrupted for life events such as child-rearing or personal or family illness or injury.

Train Staff to Build a Collaborative and Engaging Workplace

A commitment to teamwork and excellence begins with increasing cultural competency among staff, which helps develop an environment in which everyone feels welcomed, respected, and free to be their authentic selves. The Institute's Training and Leadership Development Program provides multiple professional development offerings to support this effort. These trainings are dynamic — adapting as needs are identified — and address topics such as the basics of teamwork and the role staff play in supporting a collaborative and supportive workplace. The trainings aim to promote a sense of belonging, engagement, and efficacy that is in alignment with — and reinforced by — organizational norms, values, and processes. They emphasize the crucial roles staff play in translating these norms and values into action.



RESOURCES AND INFRASTRUCTURE

The thoughtful and strategic stewardship of resources is crucial to the success of the NHLBI mission. Toward that end, rigorous oversight and guidance is applied to the sharing of data and biospecimens and to the accessibility and management of resources and infrastructure.

Strengthen Biorepositories

The NHLBI demonstrates its commitment to the responsible management of research investments by promoting scientific rigor and maximizing the use of data and biospecimen resources. We emphasize standardizing protocols and resource handling to maximize reproducibility. The Biologic Specimen and Data Repository Information Coordinating Center (BioLINCC) supports access to NHLBI-funded biospecimen and data resources and enables access to and maximizes the scientific value of both the biorepository and data repository.

Promote Data Science

The NHLBI BioData Catalyst® is an innovative data ecosystem that allows researchers to access data, analytics tools, applications, and workflows. Investigators may access data in a multidimensional environment at the scale and depth that makes the data readable for artificial intelligence and machine learning capabilities. This groundbreaking coordination of datasets promotes equal access, transparency, and collaboration between scientific partners. It also democratizes data in a way that enables reproducibility and accelerates progress toward discoveries that reduce the burden of heart, lung, and blood diseases and sleep disorders.

Operational Priorities

PARTNERSHIPS AND INNOVATION

Crucial to advancing the Institute's mission is the support, input, and insights from and collaborations with the many communities vested in HLBS research. Scientific advancement is driven through the free flow and exchange of ideas and knowledge. For the NHLBI, engagement and partnerships (within and beyond the NIH and the U.S. Department of Health and Human Services [HHS]) are foundational to the generation of scientific ideas. These exchanges occur between people with a range of expertise, influence, and interests, including patients and their families, patient advocacy groups and community organizations, local health departments, scientists, healthcare professionals, professional societies, and the media.

Enhance Partnerships

Research is a collaborative effort that would not be possible without the insights and support gained from research partners that work directly with NHLBI investigators. These have included, for example, the Bill & Melinda Gates Foundation, which supports the development of affordable, gene-based cures for sickle cell disease and HIV, and the California Institute for Regenerative Medicine (CIRM), which works with the NHLBI to explore the use of innovative technologies and therapies. Another example is our partnership with the Institute for Health Metrics and Evaluation (IHME), which supports county-level assessments of the U.S. burden of diseases and risks. Of great value are the partnerships and meaningful engagements with community-based organizations and patient representatives who work alongside our funded investigators to achieve our research objectives. These engagements provide insights that expand our perspectives on approaches to advancing HLBS research. Having effective partnerships across our portfolio is an indispensable research asset.

Promote Innovation

The Institute harnesses the creativity of individual scientists by maintaining a vigorous investigator-initiated research portfolio. The NHLBI Small Business Innovation Research/Small Business Technology Transfer (SBIR/STTR) program fosters knowledge, ideas, and innovation by supporting research and development conducted by U.S. -owned and -operated small businesses that pursue commercially promising technologies to prevent, diagnose, and treat HLBS-related diseases and disorders.

Similarly, the NHLBI Catalyze Program, a pipeline for the development of technologies and products, aims to drive the discovery of new therapies and cures; support the development of novel predictive, diagnostic, and therapeutic products; and train a diverse and globally competitive biomedical workforce. This program offers a coordinated support system to help investigators use their product development and entrepreneurial skills for innovative technologies and transformative platforms.



TRANSPARENCY AND ACCOUNTABILITY

As part of its efforts to enhance communication and engagement to build trust and confidence, the NHLBI has supported robust programs that call for community and clinic interactions that uphold key values. Those values include implementing evidence-based science and providing patients with education resources that help lead to better health outcomes. These programs are all targeted to ensure that NHLBI research reaches and benefits those who are most in need.

DISSEMINATION AND IMPLEMENTATION RESEARCH FOR POPULATIONS AND COMMUNITIES

To ensure that research progress extends to communities, the NHLBI supports dissemination and implementation research programs designed to translate evidence-based practices and interventions into real-world settings where patients and affected communities might benefit. For example, the Disparities Elimination through Coordinated Interventions to Prevent and Control Heart and Lung Disease Risk (DECIPHeR) program — an ambitious national effort — develops, tests, and supports sustainable implementation of strategies that achieve health equity in the treatment of cardiovascular and pulmonary diseases and risks while maximizing the positive impact on community health. Similarly, the Heart, Lung, and Blood Co-morbidities Implementation Models in People Living with HIV (HLB SIMPLe) program is exploring strategies for the optimal and sustainable delivery of effective prevention and treatment of HLBS comorbidities in people living with HIV in low- and middle-income countries and Small Island Developing States.



Programs like DECIPHeR and HLB-SIMPLe recognize the many factors associated with the successful implementation of evidence-based interventions and guidelines in clinical and community settings. This is especially the case for people in underserved populations — those who are part of certain racial and ethnic minority groups, who live in rural areas, and who have a lower socioeconomic status.

Educate Patients

The NHLBI supports several educational programs that aim to ensure patients receive evidence-based resources that raise their awareness about HLBS health. These resources include content related to sleep, diet, and nutrition, as well as other lifestyle issues that transcend the NHLBI portfolio. NHLBI education efforts support the advancement of evidence-based diet plans, including the Dietary Approaches to Stop Hypertension (DASH) eating plan and the Therapeutic Lifestyle Changes (TLC) program, that help improve cholesterol numbers. Other programs include The Heart Truth®, a legacy Institute program first designed to raise awareness about heart disease as the leading cause of death in women but has expanded to address heart disease as the leading cause of death for all Americans, with special emphasis on disproportionately impacted communities; the Blood Diseases & Disorders Education Program, which aims to bring greater visibility to blood donation, blood safety, and blood diseases and disorders like anemia and sickle cell disease; and Learn More Breathe Better®, which promotes awareness of lung health, including chronic obstructive pulmonary disease (COPD), asthma, and other lung diseases that affect Americans. These programs are critical to our mandate to ensure that NHLBI research progress is disseminated to meet the health information needs of health and community professionals and the public, particularly those who might benefit the most.

EMERGENCY PREPAREDNESS

By diligently establishing networks and maintaining our capacity to operate, the NHLBI can ensure that we are able to effectively respond to public health emergencies as they arise.

Establish Networks

NHLBI networks were poised to respond to the nation's research needs when the COVID-19 pandemic hit. By April 2020, the NHLBI issued a funding opportunity as part of the trans-NIH ACTIV initiative. This resulted in the launch of CONNECTS in June 2020. CONNECTS built upon NHLBI's existing clinical research networks across the nation and around the world to better understand the risk of severe illness from COVID-19. The network efficiently brought together experts from various fields, including pulmonary critical care physicians and cardiothoracic surgeons. These experts collaborated to develop a research master protocol for researching effective treatments that reduced serious illness and death in patients with COVID-19. The lessons learned from the COVID-19 pandemic will serve to ensure that the Institute remains poised to act swiftly and decisively for any future public health emergencies.

Maintain Continuity of Operations

It is crucial that the NHLBI maintains plans and resources to ensure that operations to support HLBS health continue during a crisis. Continuity of Operations (COOP) planning enables us to prepare to respond during an emergency, and it ensures the execution of essential functions through all circumstances. COOP planning is a multilayered resource that includes COOP guides and multiple training programs designed to ensure the continuous execution of essential functions through all circumstances.

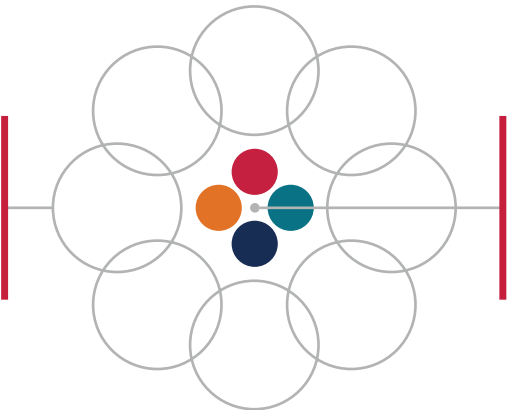
Increase the Capacity to Operate

Recognizing that today's infrastructure may not be best suited to respond to tomorrow's crises, the NHLBI is continually working to optimize our research resources so that we remain flexible and adaptable in our ability to address the next public health emergency. To that end, we are committed to diversifying our workforce, securing our research infrastructures, accelerating our research, and enriching our community relationships and partnerships.

RESEARCH PRIORITIES

The NHLBI Strategic Vision is focused on four goals, eight objectives, and five crosscutting themes: The four goals broadly define NHLBI’s vision. The eight objectives address critical challenges that create barriers to HLBS health. The five crosscutting themes address complex areas of critical need across the objectives.

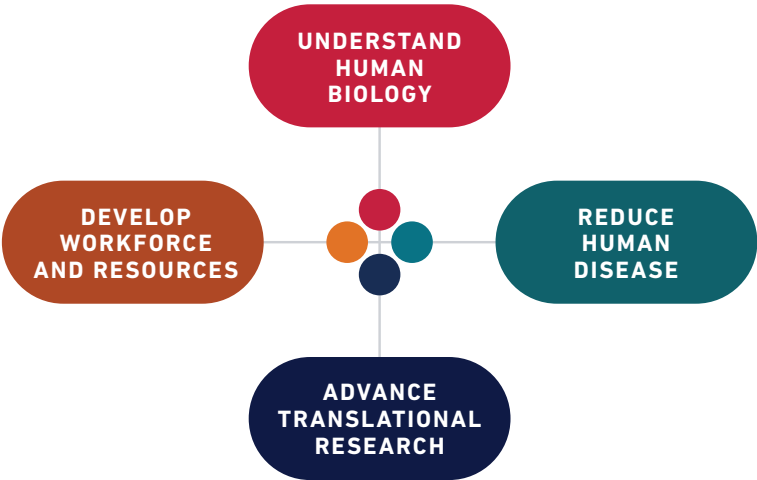
8 objectives provide a framework for moving heart, lung, blood, and sleep science forward



4 mission-oriented goals are at the center of the Strategic Vision

STRATEGIC GOALS

The four goals root the NHLBI within its mission areas and seek to understand and promote health and resilience, stimulate discoveries about the causes of disease, accelerate the translation of basic research discoveries into clinical practice, and foster training and mentoring of emerging scientists and physicians.



STRATEGIC OBJECTIVES

The eight objectives provide the framework for establishing strategic research priorities. Each objective reflects one or more of the mission-driven strategic goals and promotes the exploration of both biological and nonbiological contributors to health and disease. These objectives guide the NHLBI’s mission to propel HLBS science forward by exploring research opportunities and making strategic investments in areas ripe for discovery. Although these objectives are not meant to embody NHLBI’s entire research portfolio, they will play a key role in guiding the Institute’s ability to set, and periodically refine, a research agenda and priorities for the next decade.

STRATEGIC OBJECTIVES	
1 Understand normal biological function and resilience	5 Develop and optimize novel diagnostic and therapeutic strategies to prevent, treat, and cure heart, lung, and blood diseases and sleep disorders
2 Investigate newly discovered pathobiological mechanisms important to the onset and progression of heart, lung, and blood diseases and sleep disorders	6 Optimize clinical and implementation research to improve health and reduce disease
3 Investigate factors that account for differences in health status among populations	7 Leverage emerging opportunities in data science to open new frontiers in research on heart, lung, and blood diseases and sleep disorders
4 Identify factors that account for individual differences in pathobiology and in responses to treatments	8 Further develop and sustain a scientific workforce capable of accomplishing the NHLBI’s mission

CROSSCUTTING THEMES

The five crosscutting themes span all eight objectives and represent complex scientific challenges that require strategic support to create significant biomedical advances. They are represented by the icons below and address data science, health disparities, lifestyle behaviors, community and patient engaged research, and women’s health.



Harnessing **data science** and **new technologies** to drive scientific discovery and precision health.



Using novel approaches for **addressing health disparities** and tackling their biological underpinnings for heart, lung, and blood diseases and sleep disorders.



Furthering the science on the **importance of lifestyle** behaviors.



Leveraging the power of **community and patient engagement**.



Supporting **women's health** through the lifespan.

COMPELLING QUESTIONS AND CRITICAL CHALLENGES

Compelling questions and critical challenges are conceptual tools that help us more deeply explore challenges and opportunities related to the goals, objectives, and crosscutting themes.

- Compelling Questions are unanswered questions or poorly understood areas of research that require broad NHLBI support because their complexity exceeds the capacity of any one investigator-initiated program.
- Critical Challenges are barriers or impediments to scientific progress that, if overcome, will result in significant advances in biomedical knowledge.

There are 139 Compelling Questions and Critical Challenges outlined in the Strategic Vision. Approximately 110 are adjusted from the 2016 NHLBI Strategic Vision and 29 are new areas of inquiry.

In the pages that follow, we dive more deeply into the eight objectives, explain their importance, and propose how pursuing them has the potential to reduce the public health burden of HLBS diseases and disorders.



OBJECTIVE 1 • Understand normal biological function and resilience

Understanding normal biology is the backbone of all biomedical science. It is essential to understanding how the internal body keeps itself stable, predicting how biological systems respond to their environment, and recognizing disease pathways and targets for intervention. Research on normal biological function — including topics such as circadian rhythms, the microbiome, and tissue development from progenitor cells — can help us better define health and understand the earliest origins of disease processes. The scope of research on normal biology ranges from single cell analytics to studies of entire healthy populations. Using gold standard and emerging tools, methodologies, and technologies, such research can uncover the biological factors, behaviors, lifestyle factors, social circumstances, and environmental exposures that help people become resilient, which they must be to stay well as they age, experience stress, and endure adversity. Gaining new knowledge about the body's intrinsic capacity to repair itself will yield greater insight into the transition from health to disease. For these reasons, understanding normal biology must remain a cornerstone of NHLBI-funded research.

ENVISION A FUTURE IN WHICH WE CAN

- Promote resilience and healthy aging through the application of regenerative strategies using cells (e.g., progenitor cells) that enable the repair of HLBS systems.
- Characterize the molecular signatures of healthy heart, lung, and blood systems across the lifespan, and develop early warning systems for molecular perturbations that lead to disease.
- Pursue integrative analysis of large-scale data to characterize the molecular signature of the heart, blood, lung, and blood systems that sustain health and wellness. This data can include information derived from longitudinal electronic health records [EHR], imaging and functional tests, and multiomics by emerging systems biology, artificial intelligence, and machine learning.
- Understand how weather, and environmental exposures, social, environmental exposures, social determinants, behaviors (e.g., diet and physical activity), and genetic factors modulate biological systems, such as the epigenome, microbiome, and immune systems, to sustain health and promote resilience.



COMPELLING QUESTIONS

1.CQ.01

How are normal cell functions regulated by complex gene networks and cell-to-cell interactions?

1.CQ.02

What are the key molecular and structural mechanisms that allow single cells and tissues to sense, integrate, and respond to mechanical cues and influences at local and systemic levels?

1.CQ.03

What are the molecular, developmental, hormonal, and behavioral mechanisms and psychological, social, and environmental factors — evaluated with a systems biology integrative approach — involved in regulating HLBS function and maintaining healthy weight across the lifespan?

1.CQ.04

What are the mechanisms and range of normal physiologic responses to environmental, neuropsychiatric, social, and other stimuli that predict homeostatic resilience or transition to HLBS diseases across the lifespan?

1.CQ.05

What innate and adaptive immune system mechanisms promote HLBS health and prevent the development of HLBS diseases?

1.CQ.06

How do specific lymphatic immune and nonimmune circulatory functions interact with and contribute to HLBS health, resilience, and disease?

1.CQ.07

What is the influence of the microbiome (including virome and fungome) on HLBS health and resilience, including developmental processes, across the lifespan?

1.CQ.08

What are the pathways underlying the effects of circadian function, synchronization, and harmonization on HLBS health and resilience across the lifespan?

1.CQ.09

Does circadian regulation modify the effects of environmental exposures (e.g., cigarette smoke, vaping, particulates, pathogens, temperature, humidity) on mechanisms of HLBS function?

**COMPELLING QUESTIONS****1.CQ.010**

What are the mechanisms that underlie adaptation in HLBS systems due to place, extreme conditions, occupational exposures, weather patterns, and how can this knowledge be used to develop novel interventions that optimize health or prevent disease?

1.CQ.11

What are the basic mechanisms that direct the interactions of blood cells with each other and their environment, how do these interactions influence their function, and how can this understanding be used to optimize the handling of blood cells?

1.CQ.12

What is the normal molecular and cellular heterogeneity in HLBS health and what indicates the transition to disease?

1.CQ.13

What "omic" signatures describe the normal vasculome of the different vascular beds and different arteries (elastic vs. muscular) that supply HLBS tissues and organs?

1.CQ.14

What are the biological mechanisms and risks affecting resilience to HLBS conditions, and how do these mechanisms differ between women and men with the biological changes that occur across the lifespan?

CRITICAL CHALLENGES**1.CC.01**

Reliable and diverse investigational models that reflect individual characteristics, including based on variation in sex-based differences, are needed to reproduce normal functioning of HLBS systems and to reflect the activities of molecular targets in those systems and related diseases.

1.CC.02

Standardized protocols are needed to establish and maintain cultured cell lines and organoids relevant to functional studies of HLBS systems. Specifically, facilitating the availability of hard-to-culture cell lines and cells from female research subjects, expanding the number of HLBS cell lines, and improving reproducibility across studies are necessary.

1.CC.03

Further advancement and application of comprehensive single-cell biology analytics are needed to facilitate an integrated understanding of cellular diversity, cell interactions, and cellular phenomena in HLBS health and disease risk.



CRITICAL CHALLENGES

1.CC.04

Advances in measurement methods and models for assessing and characterizing multiple and longitudinal exposures (e.g., pre/postnatal, environmental, dietary, social) are needed to improve research on normal biologic function and resilience in HLBS.

1.CC.05

A fundamental understanding of the roles of post-translational modifications (e.g., methylation, ubiquitination, glycosylation, etc.) in HLBS health is needed.

1.CC.06

Dietary assessment methodologies that combine objective measures and biomarkers of dietary intake are needed to identify dietary patterns and food constituents that contribute to healthy weight maintenance and to inform intervention strategies to lower cardiometabolic risks.

1.CC.07

New investigative tools and knowledge of structural and matrix biology are needed to better understand injury, regeneration, and repair of the normal (or developing) heart, lung, and blood tissues and to enable regenerative medicine.

1.CC.08

An understanding of the biological HLBS functions and resilience in the context of aging and senescence during HIV infection and exposure to antiretroviral therapy.



OBJECTIVE 2 • Investigate newly discovered pathobiological mechanisms important to the onset and progression of heart, lung, and blood diseases and sleep disorders

Discovering new pathobiological mechanisms and understanding them is critical to improving HLBS health. Such discoveries and research can potentially lead to clinical and implementation studies, which in turn can inform new therapeutic strategies and clinical practice. Important areas to focus on, with the help of emerging technologies such as molecular imaging and nanotechnology, include mechanisms involved in the pathogenesis of rare and common diseases, disease-related structural and functional changes, and the clinical significance of these changes. In addition, tracking disease onset and progression across the lifespan is essential because it has the potential to inform our understanding of how chronic conditions become acute disease and how early exposures (environmental, etc.) and interventions, including lifestyle changes, impact health.

ENVISION A FUTURE IN WHICH WE CAN

- Preempt the transition from health to disease and enable disease prevention through an enhanced understanding of the pathobiological mechanisms underpinning the early onset of heart, lung, and blood diseases and sleep disorders.
- Leverage high-throughput multiomics and other advanced technologies for deep phenotyping in longitudinal cohort studies to define the functional significance of genetic and epigenetic variants on human pathobiology, as well as their impact on clinical outcomes involving heart, lung, and blood diseases and sleep disorders across the lifespan.
- Gain new insights into modifier genes and molecular pathways that influence the variable clinical manifestations and outcomes of monogenic disorders (e.g., sickle cell disease and cystic fibrosis) and structural defects (e.g., congenital heart disease), thereby catalyzing new tailored therapeutic strategies.
- Identify commonalities underlying disparate heart, lung, and blood diseases and sleep disorders to understand shared pathways and to identify shared therapeutic targets.
- Leverage emerging human complex tissue and three-dimensional cellular systems to model the pathobiological processes and to identify targetable molecular/cellular pathways that drive the onset and progression of heart, lung, and blood diseases and sleep disorders.



COMPELLING QUESTIONS

2.CQ.01

What biomarkers of acute and chronic environmental exposures are predictive of disease onset or progression? What biologic effects measured by real-time monitoring by these biomarkers are irreversible responses and which are opportunities for intervention?

2.CQ.02

How do endogenous stem/progenitor cells and defects in these cells contribute to the onset and progression of chronic HLBS diseases?

2.CQ.03

What is the pathobiology of aberrant calcification of coronary arteries, heart valves, and peripheral arteries, and why is calcification associated with a poor prognosis in HLBS diseases?

2.CQ.04

What interdependencies between the brain/peripheral nervous system and the HLBS systems are important to the development, progression, manifestation, and treatment of HLBS diseases?

2.CQ.05

What are the mechanisms whereby social conditions and psychosocial stress contributing to the onset, progression, and morbidity of HLBS systems?

2.CQ.06

Interdependencies of heart and lung disorders; How do existing lung disorders affect CVS morbidity and how does heart disease affect lung function and disease susceptibility (e.g., mechanisms whereby congestive heart failure causes lung remodeling and leads, in end-stage disease, to right ventricular failure)?

2.CQ.07

What pathobiology underlies vascular causes of cognitive decline? What early interventions could target this pathobiology to maintain cognitive function?

2.CQ.08

What is the pathobiology of fibrosis that accounts for its organ specificity (often affecting the lungs, heart, or bone marrow alone), its progression in the absence of apparent stimuli, and its resistance to drug therapy?

2.CQ.09

What is the pathophysiology of heart failure with preserved ejection fraction (HFpEF), and how can this condition be better diagnosed and treated?



COMPELLING QUESTIONS

2.CQ.10

What are the pathobiological mechanisms underlying HLBS-related pregnancy complications? What are the diagnostic, predictive, or therapeutic biomarkers for these conditions?

2.CQ.11

How can a better understanding of the molecular and physiological mechanisms of hypothermia help differentiate beneficial hypothermia from uncontrolled shock-induced hypothermia?

CRITICAL CHALLENGES

2.CC.01

Development and application of comprehensive single-cell biology analytics are needed to facilitate an integrated understanding of cellular diversity, cell-cell interactions, and cellular phenomena in HLBS health and disease risk.



OBJECTIVE 3 • Investigate factors that account for differences in health status among populations

Variations exist between populations — grouped by factors such as age, sex, race, and ancestry— in susceptibility and resilience to heart, lung, and blood diseases and disorders as well as in disease course and outcomes. Although some of these variations are caused by genetic and other biological factors, a wide range of environmental and behavioral factors and socioeconomic inequities also contribute to health disparities. Research is needed to better understand the causes of population health differences and to identify strategies to effectively address them. Investigations in this area may range from basic laboratory studies to environmental and population science to community-centered implementation research.

ENVISION A FUTURE IN WHICH WE CAN

- Leverage a deeper understanding of biologic differences related to sex or ancestral groups (e.g., variation in ancestry leading to differences in asthma phenotypes among ethnic subgroups) to devise more precise, targeted intervention strategies and to further improve clinical outcomes.
- Leverage epidemiology and the power of data science, artificial intelligence, and machine learning to understand and reduce sex-, race-, ethnicity-, and location-related health disparities.
- Incorporate measures of diet and nutrition inequities, and health disparities in research on heart, lung, and blood disease and sleep disorders across the lifespan, using existing longitudinal cohorts, innovative designs for collecting exposure data, big data approaches, individual biospecimens, and imaging using remote digital technologies.



COMPELLING QUESTIONS

3.CQ.01

How can community-engaged, late-stage translational research reduce the time needed to increase the adaptation, adoption, scale-up, and sustainment of evidence-based interventions that address the local context (including social, structural, legal, and environmental factors) in communities that experience inequities in HLBS health?

3.CQ.02

How can we improve the representation in clinical studies of individuals from all communities, and effectively engage communities as research partners to ensure that findings are relevant to everyone?

3.CQ.03

What are the determinants (environmental, structural, social, genetic, and epigenetic) of the inequities in HLBS health and the behavioral, molecular, cellular, and systemic mechanisms that result in these differences?

3.CQ.04

What factors render individuals or populations subjected to the same exposures (e.g., lifestyle behaviors, environment, psychosocial) more resilient or susceptible to HLBS diseases across the lifespan?

3.CQ.05

What are the critical lifestyle behaviors among diverse populations that affect HLBS health and the potential differential impacts of these behaviors on HLBS health?

3.CQ.06

How can we develop, test, and implement culturally appropriate and community-informed effective interventions that will have a substantial impact on HLBS health in diverse populations?

3.CQ.07

With the goal of identifying targets and interventions to mitigate inequities, what are the upstream structural factors (social, economic and legal forces, systems, and policies) and historical influences that contribute to inequities in the prevention, management, and treatment of HLBS conditions?



CRITICAL CHALLENGES

3.CC.01

Novel experimental methods and tools are needed to evaluate the effect of biological sex differences across the lifespan on HLBS health, resilience, and diseases.

3.CC.02

Precision prevention (i.e., personalized prevention within community needs) of HLBS conditions and achieving health equity requires interdisciplinary teams, advanced data science tools and analytics, diverse data sources (e.g., exposome, genetic, epigenetic, “omic”), and an inclusive research environment.

3.CC.03

Advances in methods and models for assessing and characterizing exposures (e.g., lifestyle behaviors, environmental, social exposures) and the differential effects of exposures over the life course are needed to understand the differences in HLBS health among populations.

3.CC.04

Syndemic framework-based, collaborative/multidisciplinary research is needed on HIV-associated comorbidities, coinfections, and complications in HLBS, including how social, cultural, economic, political, behavioral, and other factors, create the conditions that lead to the clustering of HLBS conditions among people with HIV, and how these factors differentially affect people with HIV (e.g., urban vs. rural, low vs. high socioeconomic status).

3.CC.05

Build trust-based and respectful partnerships with various communities that experience disparities in health to enhance inclusive and diverse participation in clinical research on HLBS disorders.

3.CC.06

Novel approaches that facilitate the engagement of underrepresented populations in all stages of research, particularly when planning interventions are critically needed.

3.CC.07

To understand the collective impact of place (including climate and environmental factors such as air quality), structural factors, and health behaviors on HLBS inequities, advanced data methods and analytical tools are needed, including relevant short-term outcome measures that predict the long-term impact on these entrenched inequities.

3.CC.08

Studies and systems that modify disparity-driving SDOH, including structural factors and barriers in health research and healthcare systems need to be designed, conducted and implemented. And their subsequent impact on health measured.



OBJECTIVE 4 • Identify factors that account for individual differences in pathobiology and in response to treatments

Research advances in areas such as genomics and other “omics” (e.g., proteomics, metabolomics) have provided new opportunities to deepen our understanding of HLBS pathobiological processes and how they vary among individual patients by sex and other factors. Accelerating progress toward precise, individualized prevention efforts and medical interventions will require research into biological factors, environmental exposures, behavioral factors, and other influences that account for differences in pathobiology and unique responses to treatment (including drug reactions and other adverse events). This research will allow personal and clinical decisions, practices, and medical products to be tailored to the individual patient to help produce the best possible outcomes.

ENVISION A FUTURE IN WHICH WE CAN

- Develop precise clinical interventions across the lifespan by sex and based on an individual’s environmental exposures, behaviors, genotypes, and molecular-cellular phenotypes.
- Accelerate the incorporation of new imaging, omics, and sensor technologies with advanced data analytics to inform a more precise clinical classification of chronic diseases and acute conditions (e.g., asthma, pulmonary fibrosis, chronic obstructive lung disease, heart failure, myocardial infarction), and enable more accurate diagnosis and treatment.
- Develop and define individual HLBS health signatures by integrating all reliable and available data (e.g., genome-wide association studies (GWAS), multiomics, genetic and non-genetic information, social/environmental factors) to guide personalized prevention strategies and clinical interventions.



COMPELLING QUESTIONS

4.CQ.01

Which phenotypic, biomarker, and molecular characteristics predict outcomes and, when applied in clinical studies, predict differential responses to therapy in individuals and in different populations with HLBS diseases? A specific focus area is the functional assays to validate millions of genetic variants to identify novel drug targets via understanding the biological mechanisms underlying differential HLBS diseases and to develop simple/affordable diagnostic tools to be deployed in outpatient settings.

4.CQ.02

What factors render individuals or populations subjected to the same exposures (e.g., diet, smoking, other environmental and social exposures) resilient or susceptible to disease? This needs to include mental exposures in the form of psychological and social stress and internal exposures in microbiome. Sex, race, and lifespan events must be considered to understand individuals' responses to the exposures.

4.CQ.03

What underlies secondary resilience, such that some people are protected from the complications of HLBS diseases? Since the secondary resilience changes over an individual's lifespan, high-impact life events must be considered, such as menopausal transition and its impact on cardiopulmonary health and sleep disorders.

4.CQ.04

Which patients benefit from rehabilitation treatments (e.g., cardiac, vascular, pulmonary), and how can the benefits of rehabilitation treatments be sustained long term? The impact of environmental and omics profiles should be considered, and special attention should also focus on sex and race differences in responding to the treatment.

4.CQ.05

How does the pathobiology that underlies nonobstructive ischemic heart disease and the associated risks for acute coronary syndrome and early mortality differ between subpopulations, and what are the targets for treatment and prevention?

4.CQ.06

What tests would identify individuals who are at high risk of venous thromboembolic events and who would benefit from targeted risk factor modification and/or intensive prophylaxis?

4.CQ.07

What are biomarkers of pulmonary hypertension that could better identify individuals at high risk, reveal underlying mechanisms, and guide treatment? For instance, considering pulmonary arterial hypertension (PAH) predominantly affects women, what sex-specific risks (e.g., rurality, conventional, SDOH, mental health, societal constraints, menopausal status, adverse pregnancy outcomes) across the lifespan enable epigenetic analyses to understand susceptibility and pathways for prevention, resilience, or successful aging.

4.CQ.08

What are the major determinants of individual and sex differences in breathing patterns in sleep, susceptibility to insomnia, and other sleep behaviors?



COMPELLING QUESTIONS

4.CQ.09

What genetic biomarkers, genetic comparisons across ethnic groups, and environmental predictors of sickle cell disease risk would help inform and improve management of the disease's progression and its complications.

CRITICAL CHALLENGES

4.CC.01

Predictive modeling and prevention trials are needed in populations at high risk for highly prevalent HLBS diseases. The advancement in environmental and omics profiles of HLBS diseases should be leveraged to enable early prediction/prevention for high-risk individuals and to minimize trial and errors during the development of therapeutic treatment of HLBS diseases.

4.CC.02

Greater use of advanced technologies in precision medicine research is needed to accelerate investigations of HLBS disease mechanisms and the development of targeted treatments for complex HLBS diseases.

4.CC.03

Clinical evaluation tools are needed to differentiate patients with atherosclerotic heart disease who will progress to myocardial infarction or with sudden cardiac death from those with stable disease.

4.CC.04

Portable, wearable technologies tailored for women that capture real-time, continuous individual data with minimal interference, need to be promoted. The data from these technologies should be objectively surveilled and should capture critical risks of HLBS diseases, such as blood pressure and blood sugar, arrhythmias, asthma/COPD exacerbation, behaviors, and adverse stressors. Such data would enable investigations of risks and adverse maternal outcomes.

4.CC.05

Ethical, legal, and social implications of patient privacy and data security, racial and cultural factors, and diversity in precision medicine research need to be considered.



OBJECTIVE 5 • Develop and optimize novel diagnostic and therapeutic strategies to prevent, treat, and cure heart, lung, and blood diseases and sleep disorders

Recent scientific and technological developments offer especially promising opportunities to prevent disease, manage and treat illness, and promote resilience. These wide-ranging developments include new gene editing techniques to safely treat and prevent HLBS diseases and disorders, smart tools that monitor and adjust biological processes, techniques that modulate stem cell or immune system signaling to reduce disease risk, and advances in materials science that could yield vastly improved implant devices. Other technologies and bioengineering developments may further facilitate diagnostic capabilities. New lifestyle interventions may improve the maintenance of continued health and wellness and may enable behavioral modification to prevent disease. Many of these advances, which are the results of past investments in basic research, are crucial to developing and optimizing novel diagnostics and therapeutics strategies.

ENVISION A FUTURE IN WHICH WE CAN

- Develop surgical grafts (using stem cells, 3D printing, and nanotechnologies) for children born with heart defects, in which the grafts grow and adapt into adulthood, making multiple operations unnecessary.
- Transform the safety and availability of the nation's blood supply by using genetically engineered stem cells to eliminate histoincompatible antigens and generate universal donor red blood cells that are available off the shelf by a high-throughput process.
- Build intelligent health systems for HLBS diseases and disorders that use artificial intelligence and machine learning tools that can screen patients for risk factors for early disease development and that can diagnose and evaluate the effects of interventions and treatments.
- Create biomimetic platforms that promote lung tissue regrowth, regeneration, repair, and replacement of a patient's individual lung cells and/or tissues to reduce the burden of underlying lung diseases.



COMPELLING QUESTIONS

5.CQ.01

Would reduction of known cardiac and vascular risk factors during childhood and adolescence translate into the prevention or delayed development of atherosclerosis and other heart diseases?

5.CQ.02

Would interventions in pregnancy or early childhood designed to modulate immune development result in primary prevention of asthma?

5.CQ.03

How should the management of diseases that typically develop in childhood (including childhood interstitial lung disease, hemoglobinopathies, congenital heart disease, cystic fibrosis, and asthma) be modified as affected individuals mature into adulthood?

5.CQ.04

How does applying lifestyle interventions (diet, nutrition, exercise) as a part of routine patient care in a variety of contexts—from community to patient care settings—help manage or prevent lung function decline in the general population and patients with chronic lung diseases?

5.CQ.05

How do we apply circadian-based strategies (e.g., sleep, timing of medication, meals, device monitoring) to improve the efficacy of treatments for HLBS diseases (e.g., hypertension, asthma, thrombosis, obesity/diabetes)?

5.CQ.06

What technical improvements in the collection, preparation, storage, and processing of blood and synthetic blood products would improve their potency, safety, and lifespan? What animal model improvements are needed for hemoglobin-based oxygen carriers (HBOC) studies to accelerate improvements in blood products?

5.CQ.07

What effective and implementable practices (e.g., recognition and initial response by the community, emergency medical response, hospital-based care) would reduce the rate of mortality associated with out-of-hospital cardiac arrest?

5.CQ.08

What new interventions or surgical tools are needed to achieve, with minimal symptoms, the optimal clinical management for patients with severe calcific aortic stenosis?

5.CQ.09

In patients with enhanced cardiovascular risk due to comorbidities from chronic diseases (e.g., HLBS disorders, diabetes) and multiple drug therapies (including cancer survivors), what new biomarkers are needed to help predict at-risk patients, improve outcomes, and help guide treatment management?



COMPELLING QUESTIONS

5.CQ.10

What are the underlying risk factors to predict early complications versus resilience after hematopoietic stem cell transplantation (HSCT)? What are the mechanisms for efficiency of differentiation of hematopoietic stem cells (HSCs)? How can these consequences be predicted and prevented to reduce the high rates of mortality following HSCT?

5.CQ.11

How can we “reprogram” the immune system to improve outcomes of allogeneic cell therapies, tissue and organ transplants, and regenerative strategies and to diminish allogeneic responses to essential biologic replacement therapies?

5.CQ.12

How do we develop and implement novel strategies to prevent and treat minor and major hemorrhagic complications in individuals affected by acquired and inherited disorders?

5.CQ.13

How can targeted manipulation of epigenetic modifications (distinct from global suppression of histone acetylation or DNA methylation) become a useful strategy for therapeutic intervention in chronic cardiopulmonary or blood diseases?

5.CQ.14

Do interventions to improve ventilation during sleep decrease morbidity and mortality in individuals with either heart failure (or other diseases associated with chronic hypoxemia) and sleep-disordered breathing?

5.CQ.15

How can alterations of stem cell cycles and other therapies, as well as endogenous mechanisms, be harnessed to promote repair and regeneration of the heart, lung, and blood systems?

5.CQ.16

How can we better integrate palliative care concepts, such as respect for personal values, goals, and treatment preferences, in the management of patients with HLBS diseases?

5.CQ.17

Can strategies targeting lung and airway progenitor cells be used for early diagnosis, prevention, and treatment of chronic obstructive lung diseases in high-risk populations?



CRITICAL CHALLENGES

5.CC.01

An understanding of the immune system from a systems biology perspective is needed to design more efficacious treatment strategies for chronic inflammatory and autoimmune HLBS diseases.

5.CC.02

Improved capabilities for responding rapidly and effectively to emerging infectious threats to the safety and availability of the nation's blood supply are needed.

5.CC.03

Robust tools and algorithms are needed to evaluate objective biomarkers of sleep health and dysfunction.

5.CC.04

New materials and constructs that are electrically, chemically, and mechanically active are needed to enable the development of self-adjusting bioengineered implants (e.g., synthetic/biomimetic tools, bioprinted tissues relevant to HLBS).

5.CC.05

Development of safe, well-functioning designer platelets and red blood cells from stem or progenitor cells, as well as the large-scale production of these products, is needed for therapeutic and diagnostic uses.

5.CC.06

A variety of smart devices are needed that both monitor physiology and integrate data from the digital health devices into clinical decision-making and empower patients' self-care and HLBS disease management.

5.CC.07

Optimization of ventricular assist devices for HLBS to incorporate better percutaneous and transcutaneous systems and to improve battery and charging-mechanism designs is needed.

5.CC.08

A more rapid translation of new discoveries about molecular, cellular, and tissue-based mechanisms of better therapeutic and preventive strategies for HLBS diseases is needed.

5.CC.09

Understanding overlapping etiology and pathogenesis, such as chronic inflammation, immune dysfunction and aging, and microbiome/virome contributions, that promote HIV-associated comorbidities and complications affecting HLBS systems to enable the development of novel interventions for the prevention, control, and treatment of such comorbidities and complications is needed.



CRITICAL CHALLENGES	
5.CC.10	Safer and more efficacious hematopoietic stem cell and gene therapy approaches to eliminate HIV reservoir, cure HIV infection, and improve immune system restoration are needed.
5.CC.11	Improved infrastructure and workforce expertise in HLBS-related biomanufacturing is needed.
5.CC.12	Comprehensive analysis of clinical, imaging, and multiomics data using artificial intelligence and machine learning needs to be leveraged to identify disease endotypes and to develop endotype-specific personalized disease-modifying interventions for HLBS disorders.
5.CC.13	Comprehensive analysis of multiomics/bioinformatics and advanced imaging data using artificial intelligence and machine learning need to be leveraged to discover the biomarkers of the onset of chronic obstructive lung diseases and cardiovascular diseases in smokers and to develop strategies that prevent disease onset and progress.



OBJECTIVE 6 • Optimize clinical and implementation research to improve health and reduce disease

For maximum return on NHLBI investments in research and research training, it is important to prioritize strategies that support integration of evidence-based interventions within clinical and public health settings. New methodologies, research frameworks, and resources must be developed to refine the conduct of clinical studies and implementation research, while also improving the overall efficiency, effectiveness, and impact of the research enterprise. Current challenges include recruiting research participants of all backgrounds, effectively research participants, effectively translating knowledge gained from these studies into programs and policies that improve health at the clinic and community levels, and engaging in a strategic, meaningful, and sustainable way with community partners at every stage of the research process to build trust and enhance the impact of this research. Commitment to addressing these challenges is paramount, as is the careful attention to building and nurturing meaningful and sustainable community-engaged research partnerships, especially in communities disproportionately burdened by health disparities. Also important are multi-, inter-, and transdisciplinary collaborations that make the most of clinical and implementation research efforts.

ENVISION A FUTURE IN WHICH WE CAN

- Easily and safely link EHRs and personal health data (e.g., from patient-reported outcomes and wearables) to help streamline clinical research and empower research volunteers to self-identify.
- Accelerate the translation of discovery into practice by leveraging HLBS research platforms (e.g., clinical epidemiology registries, clinical trial networks, implementation science programs) to engage the clinical care and patient communities; leverage the use of EHRs, patient registries, and existing datasets; and create a seamless interface between clinical research and practice.
- Build sustainable community partnerships to support research and advance health equity.



ENVISION A FUTURE IN WHICH WE CAN

- Implement strategies for widespread access (e.g., in rural settings and at safety net hospitals, primary care facilities, and community health centers) to real-time health education and to prevention and disease monitoring tools, using telemedicine and remote monitoring devices that help improve health outcomes for individuals with HLBS diseases and disorders.
- Design scalable strategies for integrating and sustaining proven-effective interventions in communities where they are needed the most.

COMPELLING QUESTIONS

6.CQ.01

What approaches (methods, such as community engagement, or technologies) are effective for raising awareness and participation in clinical research across diverse population groups and across the lifespan, and how can we incorporate these approaches in clinical trial design?

6.CQ.02

What clinical trial designs are best to investigate factors responsible for variation in individual responses to pharmacological and other interventions?

6.CQ.03

How can we engage all interested, affected, and/or relevant parties (e.g., patients, providers, health systems, regulatory bodies) to improve the clinical research enterprise by addressing critical regulatory, operational, and fiscal needs to enhance efficiency and make evidence-based interventions equitably accessible and affordable?

6.CQ.04

How can we use dissemination and implementation science to identify, develop, and test strategies to improve awareness and access to evidence-based diagnostics, prevention practices, and therapeutics, including emerging approaches to care, to speed up, and to maximize generalizable knowledge and use of effective strategies?

**CRITICAL CHALLENGES****6.CC.01**

Meaningful, diverse, and inclusive collaboration and engagement among researchers from various backgrounds and all academic levels and disciplines (e.g., basic science, translational scientists, implementation scientists; clinicians; population scientists, engineers, bioinformatics experts, data scientists) and sectors (e.g., nonprofit organizations, industry), including community members, are needed to enhance and expedite advances in HLBS research and turn discovery into health.

6.CC.02

Skills development and training, as well as new methods, are needed to improve the navigation of preclinical new product phases within the context of the entire clinical translational science continuum to speed the pipeline from discovery to impact.

6.CC.03

Expanded resources for identifying therapeutic targets and agents, establishing proof of concept, and developing data for investigational new product applications are needed across the lifespan to enable the early translation of research findings to clinical applications.

6.CC.04

Creative approaches to clinical trials in rare HLBS diseases are needed across the lifespan to successfully test strategies that will expand preventive and therapeutic options.

6.CC.05

Novel methodologies and improvements in existing methodologies are needed for dissemination and implementation research that explores uptake of research findings into approaches, programs, and policies to yield practical, actionable evidence.

6.CC.06

Multidisciplinary, multinational, multilevel partnerships are needed to develop effective and sustainable strategies for combating chronic HLBS disorders in low- and middle-income countries, which take into account the highly variable local epidemiology of HLBS disorders, the need for novel approaches to reducing disease burden, and the implementation challenges inherent in under-resourced settings.

6.CC.07

Increased support is needed for implementation and dissemination research to facilitate the adaption and scale-up of proven interventions in community and clinical settings for the prevention, control, and treatment of HIV-associated HLBS comorbidities, coinfections, and complications across the lifespan, especially among vulnerable and underserved populations who often have limited access to care.

6.CC.08

Insights on costs, resources to support partnerships, and the means of financing successful dissemination and implementation strategies are needed to enable the understanding of short-term and long-term investments necessary to support systemwide integration, sustainable use, and possible adaptation of successful implementation strategies.



CRITICAL CHALLENGES

6.CC.09

There is a need to engage and support a broader diversity of institutions, communities, and sectors, including the primary care sector, in clinical trials through enhanced clinical research infrastructure domestically and globally to drive inclusive participation across multiple dimensions (race, ethnicity, geography, socioeconomic status) to enhance relevance and impact of HLBS science.

6.CC.10

Enhanced efficiency, cost containment, data sharing, and scientific rigor of clinical research of living, labor, and technologies and community engaged research are needed.

6.CC.11

Clinical research infrastructure support is needed to enable the rapid implementation and conduct of clinical trials in a diverse setting (including low resource/primary care).



OBJECTIVE 7 • Leverage emerging opportunities in data science to open new frontiers in research on heart, lung, and blood diseases and sleep disorders

New technologies — from omics platforms to high-throughput screening — have generated vast amounts of data that have the potential to provide new insights into the preemption and precise treatment of HLBS diseases and disorders. Unfortunately, only a small portion of these datasets are being optimally assessed and incorporated into practice. Additional consideration must be given to the ethical, legal, and social implications of artificial intelligence use and data-sharing approaches in disease diagnosis and management. Developing innovative approaches to the integration, analysis, and interpretation of data from multiple sources will be essential. This information can then be effectively used to understand biological, social, environmental, and behavioral determinants associated with HLBS health and disease and to improve patient outcomes.

ENVISION A FUTURE IN WHICH WE CAN

- Accelerate the discovery of mediators of health and disease through the seamless and shared use of readily accessible cloud-based datasets that integrate genomics and other omics analyses with carefully phenotyped research participant data from diverse sources.
- Develop handheld devices to process and integrate genomics and clinical and personal health data at the bedside to assist clinicians in their care of patients.
- Leverage the NHLBI BioData Catalyst, NHLBI biorepositories, and other data resources to evaluate emerging therapeutics for HLBS diseases and disorders using cutting-edge data science, bioinformatics, and synthetic biology techniques.
- Integrate data resources to create a digital lung, which incorporates the molecular, cellular, pathophysiology, and physical dynamics of the lung to gain a better understanding of chronic lung disease and to develop data-driven treatment strategies.

**COMPELLING QUESTIONS****7.CQ.01**

As the research enterprise evolves to include large multimodal datasets, how can we train multidisciplinary teams about topics such as biostatistics, computer science, and bioinformatics, so they can tackle some of the most challenging and complex HLBS research questions?

CRITICAL CHALLENGES**7.CC.01**

The development, application, and sharing of robust and multidimensional data-analytical and theoretical methods, mathematical modeling, and computational simulation techniques are needed for understanding fundamental mechanisms of HLBS systems, including gene, protein, and metabolic regulatory networks, and the impact of environmental exposures on those networks.

7.CC.02

Novel integrative systems biology and analytical approaches are needed to exploit the wealth of data coming from genetics, epigenetics, transcriptomics, metabolomics, proteomics, environmental exposures, SDOH, electronic health records, and imaging to define disease subtypes, predict risks, and identify therapeutic targets.

7.CC.03

Novel analytical approaches, coordinated access to data, well-planned sample analyses, and the creation of a scientific data commons are needed to leverage existing deeply phenotyped cohorts and accelerate translational research to promote the discovery of key druggable targets and the development of novel and precise treatments for HLBS diseases.

7.CC.04

Advancements are needed in the organization, infrastructure, integration, and availability of multimodal data, including imaging, EHR, and wearable data, and omics data, including genetic, epigenetic, transcriptomic, metabolomic, proteomic, phenotypic, and ontological information.

7.CC.05

Bold new data science methods, including artificial intelligence and machine learning approaches, are needed to improve the analysis of big data, including population, trial, and cohort research.

7.CC.06

Advanced analytic methodologies should be used to efficiently and rapidly leverage existing contextual, pragmatic, administrative, and original datasets to address critical challenges such as methods, measures, generalizable knowledge, scale, and spread of effective implementation strategies.

7.CC.07

Multimodal analyses of data from a variety of sources (e.g., registry data, molecular, social, behavioral, environmental exposures, wearable sensor, self-reported data) are needed to develop predictive and actionable models for understanding the risk, prevention, and treatment of HLBS disorders.



OBJECTIVE 8 • Further develop and sustain a scientific workforce capable of accomplishing NHLBI's mission

New approaches are needed to ensure the continued development of a scientific workforce of diverse disciplines and backgrounds, equipped with relevant skills, knowledge, and resources to tackle future HLBS challenges. This goal will require strategic interventions throughout the research career continuum, which spans across K to 12 education, collegiate and postdoctoral studies, and career development from early investigator to senior scientist. Collaborative partnerships will be crucial to successfully expanding the exposure of young students to the wonders of science and sustaining their interest through college and graduate education. Embracing research training as a lifelong exploration of scientific curiosity will also remain critical, as will the NIH's continued promotion of new scientific disciplines, tools, and technologies that keep pace with an ever-changing biomedical and public health landscape.

ENVISION A FUTURE IN WHICH WE CAN

- Enable innovative solutions to pressing health problems by promoting the use of multidisciplinary teams capable of leveraging diverse skills and perspectives from crosscutting research fields such as data science, computational biology, bioengineering, quantitative population sciences, behavioral science, and implementation science.
- Build an NHLBI workforce and research portfolio that reflect the widest possible range of scientific thinking, perspectives, backgrounds, and lived experiences.
- Continually train the research workforce across all career stages in evidence-based science, cutting-edge methodologies, and emerging areas of research.
- Educate students from diverse backgrounds about biomedical research-related careers and encourage their pursuit of those careers through earlier introduction to research opportunities and scientific advancements.
- Quickly identify and address the barriers facing the growth of the next generation of researchers, particularly in lung and sleep science and the data sciences.



COMPELLING QUESTIONS

8.CQ.01

What kinds of exposures, beginning in early education, would stimulate and maintain students' understanding and interest in science, particularly students from diverse and disadvantaged backgrounds?

8.CQ.02

How can we build and sustain a diverse HLBS scientific workforce that includes people from different groups and from all regions of the county, especially those from rural, disadvantaged, and underserved communities?

8.CQ.03

How can research training programs be tailored to attract and retain researchers with cultural competency to address health challenges through a scientific lens that includes location and neighborhood- and community-level SDOH that contribute to health disparities and outcomes?

8.CQ.04

What are the best strategies to develop a scientifically diverse workforce across the full spectrum of investigative approaches — from basic to clinical, clinical to practice, and population to health implementation science — to address health inequities and disparities, including those of women's health, domestically and globally?

8.CQ.05

How do we ensure that HLBS trainees across the career continuum are aware of and prepared for a variety of possible scientific career opportunities (e.g., careers in teaching, industry, government)?

8.CQ.06

How do we best develop a scientific workforce that is fluent in product development and commercialization issues, including regulatory, intellectual property, and business issues, in order to bring products for HLBS indications to the market?

8.CQ.07

How do we attract students to, and support trainees in, traditional basic science disciplines (e.g., biochemistry, physiology, pharmacology, integrative biology, data science) in order to address research topics that are important to the NHLBI and to advance emerging fields of science?

8.CQ.08

How do we add communication skills to our training programs to improve scientists' messaging with the public so they can better promote health, disseminate scientific advances, and counter misinformation?



COMPELLING QUESTIONS

8.CQ.09

How can we harness virtual technologies to expand collaboration, foster creative mentoring, and expand access to advanced instruction and remote learning, to address the needs of the modern and future biomedical workforce in the United States and globally?

8.CQ.10

How can we better incorporate interdisciplinary and team science in our training and career development programs to prepare scientists for collaborative research and for using emerging technologies and resources? How do we also improve cross-communication between scientists in different fields to convey and appreciate the importance of each other's research for more effective teams?

8.CQ.11

How can we encourage senior scientists to collaborate with or mentor early career investigators to develop new research programs and support the independence of emerging leaders in their field?

8.CQ.12

How can community healthcare workers be better educated about scientific advances to improve care in underserved communities?

8.CQ.13

How can we increase participation of nonphysician clinicians, such as nurses, advanced practice providers, clinical psychologists, social workers, veterinarians, dentists, and others, in our training programs and research workforce as a whole?



CRITICAL CHALLENGES

8.CC.01

Sufficient numbers of clinical scientists are needed, particularly those interested in pursuing the translation of breakthroughs from basic science laboratories into clinical settings. Research training opportunities are needed for all stages of clinical training, beginning early in their professional education.

8.CC.02

Programs of training, professional development, and mentoring are needed to help create a more diverse cadre of senior leaders in science and medicine.

8.CC.03

Training that emphasizes rigorous scientific methods in the biomedical, behavioral, social, data, and implementation sciences is required to increase reliability and reproducibility of research findings.

8.CC.04

Better support is needed for scientists who are making transitions between career stages (e.g., the graduate/medical education stage, the postdoctoral/fellowship period, the junior investigator stage, first independent clinical trial for clinical investigators).

8.CC.05

Curricula and resources for education of healthcare workers in evidence-based care are needed.

8.CC.06

Improved methods for standardized collection and analysis of education, employment, and productivity data from HLBS scientists over the course of their careers are needed to assess defined metrics and predictors of success at both individual and training program levels.

8.CC.07

More early career investigators need to be recruited and supported, and the success of those interested in research on HIV and HLBS sciences needs to be facilitated.

8.CC.08

Training opportunities need to be increased for investigators across the career span in emerging areas of research, particularly in the use of artificial intelligence and machine learning, and other data science approaches to facilitate the analysis of increasingly large biomedical datasets.



CRITICAL CHALLENGES

8.CC.09

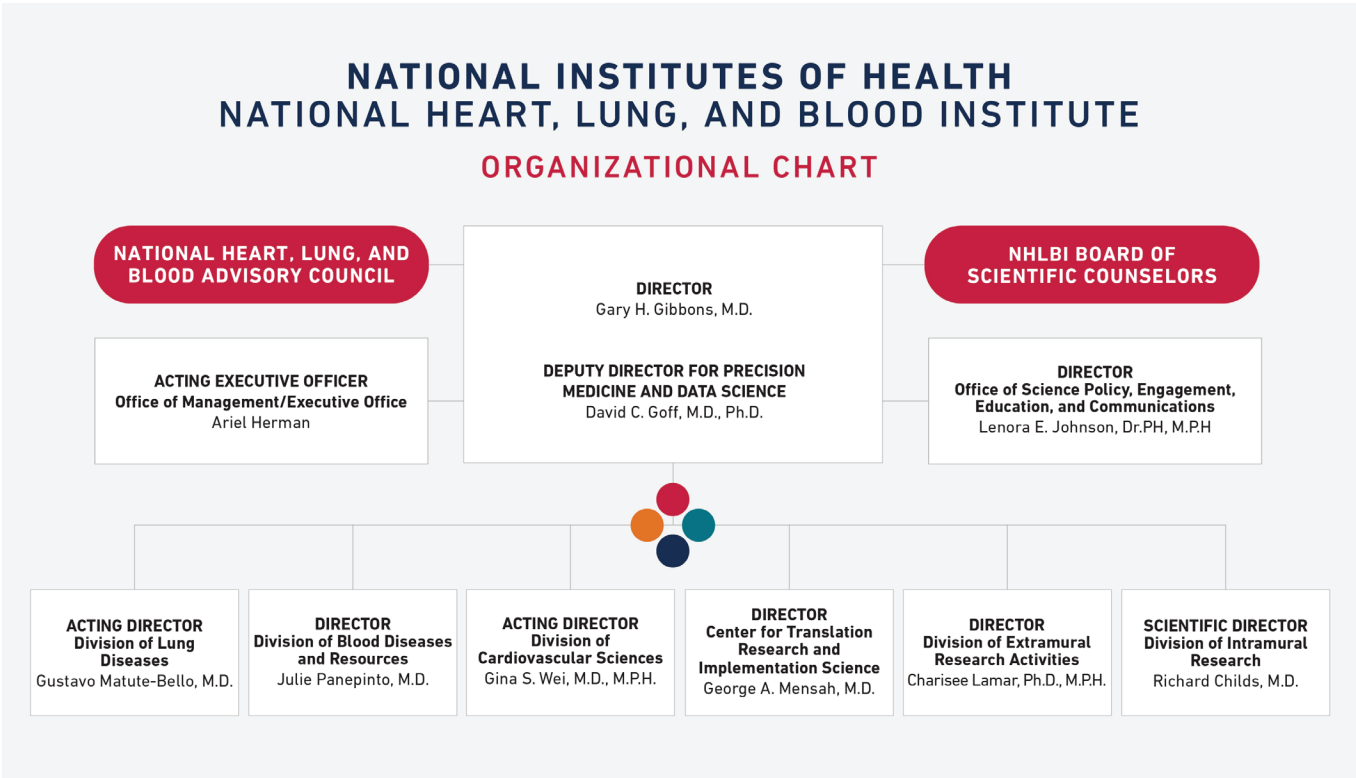
Scientists across all career/life stages, especially women who more often experience challenging and disruptive life events, need to be better supported through research training, career development, and capacity building initiatives.

8.CC.10

The pool of postdoctoral fellows who qualify for National Research Service Award (NRSA) training programs needs to be expanded.

APPENDIX I

NATIONAL INSTITUTES OF HEALTH NATIONAL HEART, LUNG, AND BLOOD INSTITUTE ORGANIZATIONAL CHART





U.S. Department of Health and Human Services
National Institutes of Health
National Heart, Lung, and Blood Institute